Computer Graphics: Seeing the Digital World

From Pixel to immersive experiences

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Introduction to Computer Graphics

- The art and science of creating,manipulating and displaying visual content using computer

Key Components

- Creation of digital visual content
- Representation and manipulation of image data
- User interfaces, visualization, and simulation

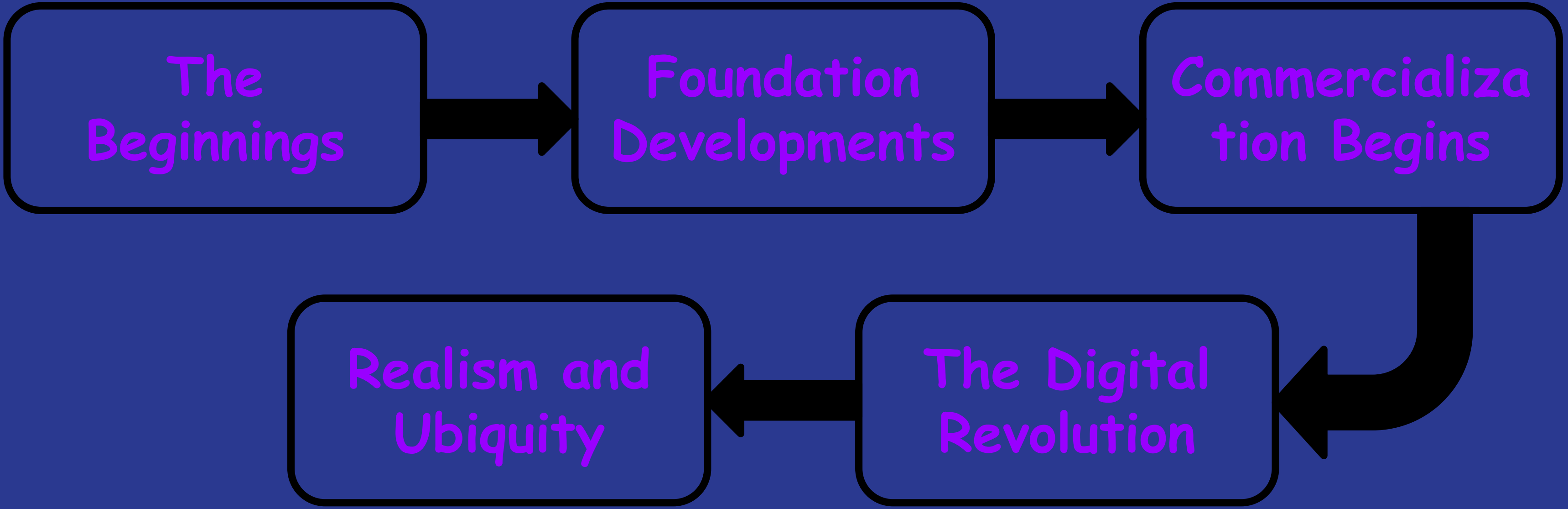
Two Main Categories

- Raster Graphics
- Vector Graphics

Categories of Computer Graphics

VECTOR GRAPHICS	RASTER IMAGES
Made of mathematical paths (lines, curves, shapes)	Made of pixels (tiny squares of color)
Common formats: AI, EPS	Common formats: JPG, PNG, TIFF
Best for logos, icons, line art text-based graphics	Best for photos, textures, and detailed graphics
Resolution independent; scales infinitely up or down	Enlarging or reducing can impact quality
Must be converted for use on websites	Works directly on websites

History of Computer Graphics



Overview of Graphics Systems

Video Display Devices

Video display devices in computer graphics are hardware, primarily monitors, that convert digital signals into visual images using technologies like Cathode-Ray Tubes (CRTs) or flat-panel displays (LCD, LED, Plasma).

1. CRT Displays (Uses an electron gun to strike a phosphor-coated screen):
    - Raster-scan displays
    - Random-scan displays
2. Flat panel displays (Thinner, lighter, and more energy-efficient alternatives to CRTs):
    - LCD
    - LED
    - Plasma Panels

3. Three-dimensional viewing devices

- 3D viewing devices provide depth perception to users which enhances visual experiences by creating the illusion of three-dimensional spaces.
- Examples:- Two-view stereoscopic or autostereoscopic displays

Components and Techniques

- Pixel: The smallest addressable element on a screen.
- Frame Buffer: Memory used to store the intensity values of all pixels for a displayed image.
- Refresh Rate: The number of times per second the image is redrawn, crucial for avoiding flicker.

Graphics software and tools

Application Programming Interfaces (APIs):

- OpenGL (Cross-platform, established standard)
- DirectX (Microsoft Windows ecosystem)
- Vulkan (Low-overhead, cross-platform)
- Metal (Apple ecosystem)
- WebGPU (Web standard, emerging)

Development Tools & Libraries:

- Game Engines: Unity, Unreal Engine, Godot
- 3D Modeling: Blender, Maya, 3ds Max
- 2D Graphics: Adobe Suite, GIMP, Inkscape
- Data Visualization: D3.js, Three.js, Matplotlib

Graphics Pipeline

- The graphics pipeline (or rendering pipeline) is a series of stages that converts 3D scene data—such as vertices, textures, and lighting—into a 2D image on a screen.

- It operates by taking CPU-driven draw calls and processing them on the GPU through stages like vertex shading, rasterization, and pixel shading.
- This process is largely parallelized, allowing for real-time rendering of complex graphics.

2D Viewing Pipeline

- a sequential series of transformations that map 2D object data from a world coordinate system to a specific viewport on a display device.

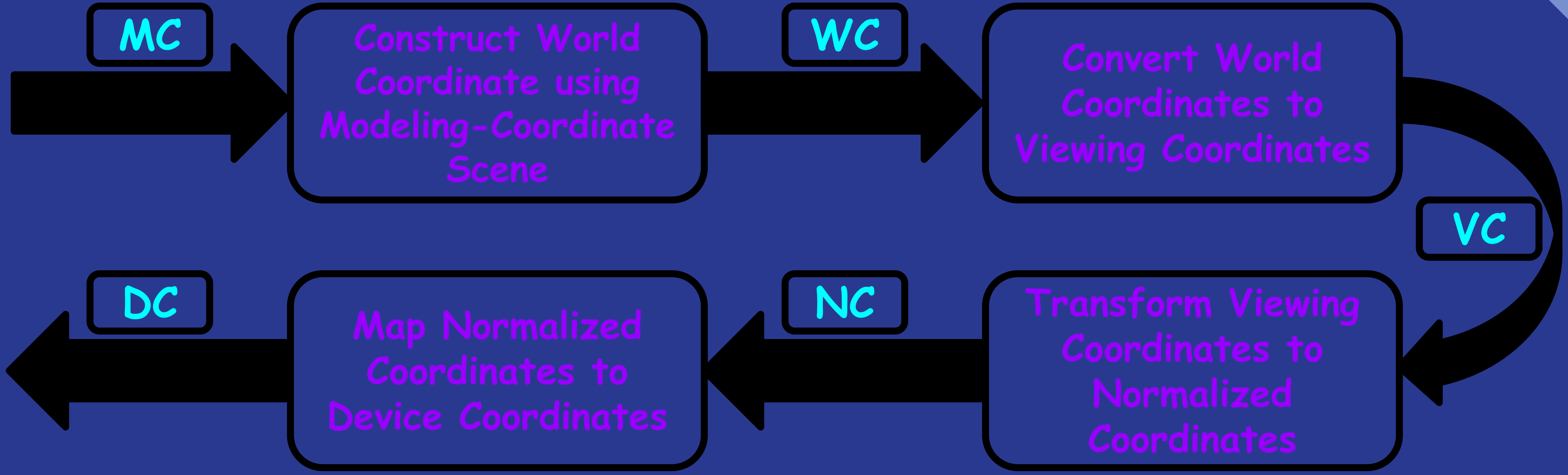


Fig:- Flowchart for 2D Viewing Pipeline

3D Viewing Pipeline

- a structured sequence of transformations that converts 3D world coordinates into a 2D image for display.

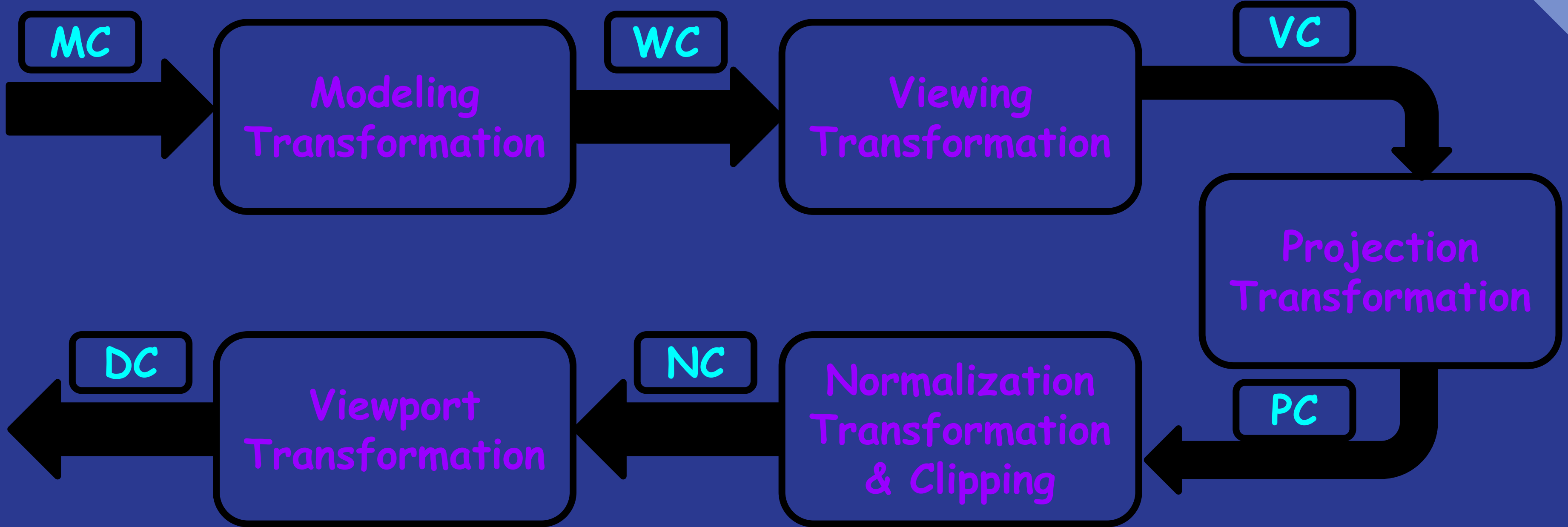


Fig:- Flowchart for 3D Viewing Pipeline

Applications of Computer Graphics

Medicine

- MRI/CT scan visualization
- Surgical planning and simulation
- Medical training

Engineering

- Automotive and aerospace design
- Architectural visualization
- Product design and prototyping
- Virtual replicas of physical systems

Augmented Realism

- Mobile AR applications
- Industrial maintenance
- Medical visualization

Virtual Realism

- Immersive training (aviation, military)
- Virtual tourism and real estate
- Therapeutic applications

Arts

- Animation and Motion Graphics
- Graphic Design and Digital Imaging
- Morphing and Special Effects
- 3D Modeling and Sculpting

Future Trends

- Real-time Ray Tracing
- AI-Assisted Graphics (DLSS, neural rendering)
- Quantum Computing for Graphics
- Holographic displays
- Brain-computer interface visualization

Conclusion & Key Takeaways

Summary:

1. Computer graphics has evolved from simple wireframes to photorealistic real-time rendering
2. Hardware advances (GPUs) have driven capability explosions
3. Applications now touch every aspect of modern life
4. The future promises even more immersive and accessible visual experiences

The Essence: Computer graphics transforms data into understanding, ideas into visuals, and imagination into reality.

Q & A and Discussion

Discussion Points:

- Ethical considerations in photorealistic CGI
- Environmental impact of graphics computing
- Accessibility in graphics tools and applications
- Career paths in computer graphics

THANK YOU!!!